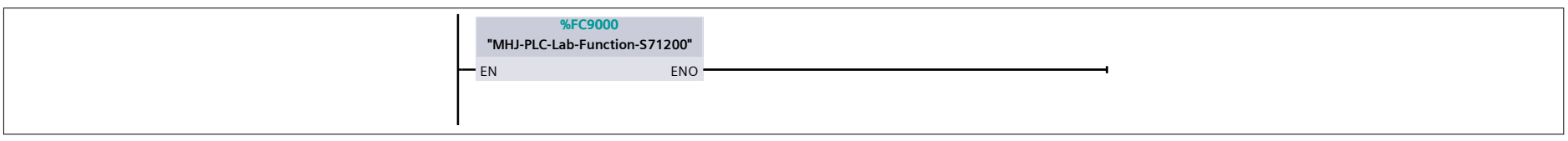
Program blocks

Main [OB1]

Main Properties							
General							
Name	Main	**Number**	1	**Type**	OB	**Language**	LAD
Numbering	automatic						
Information							
Title	"Main Program Sweep (Cycle)"	**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

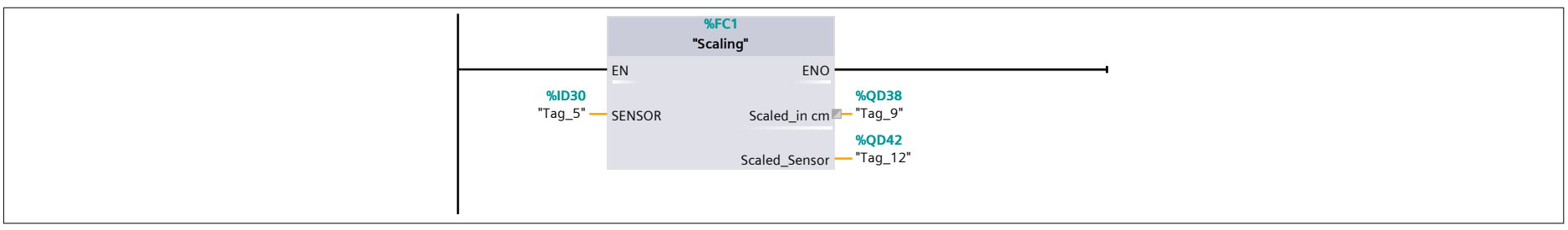
Name	Data type	Default value	Comment
▼ Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

Network 1: Code for Linking Siemens TIA and Factory IO



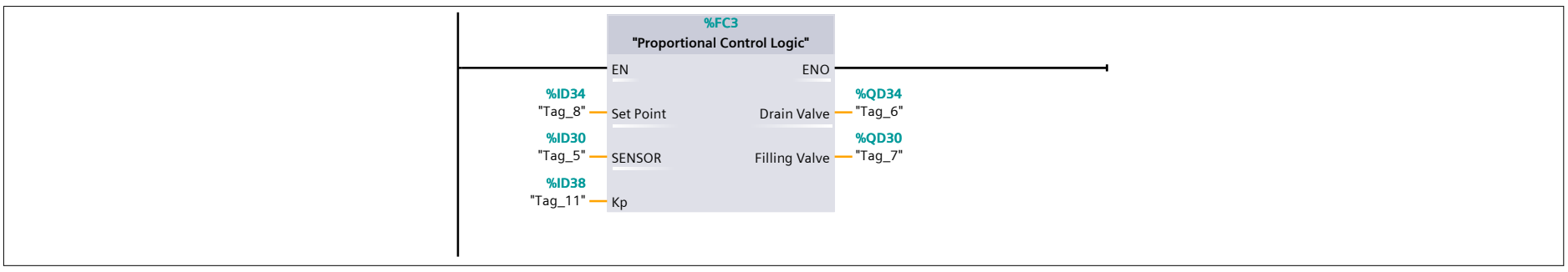
Symbol	Address	Type	Comment

Network 2:



Symbol	Address	Type	Comment
"Tag_5"	%ID30	Real	
"Tag_9"	%QD38	Real	
"Tag_12"	%QD42	Real	

Network 3:



Symbol	Address	Type	Comment
"Tag_5"	%ID30	Real	
"Tag_6"	%QD34	Real	
"Tag_7"	%QD30	Real	
"Tag_8"	%ID34	Real	
"Tag_11"	%ID38	Real	

Program blocks

MHJ-PLC-Lab-Function-S71200 [FC9000]

MHJ-PLC-Lab-Function-S71200 Properties							
General							
Name	MHJ-PLC-Lab-Function-S71200	**Number**	9000	**Type**	FC	**Language**	SCL
Numbering	manual						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Output			
InOut			
▼ Temp			
rdTimeReturn	Int		
▼ outputTime	DTL		
YEAR	UInt		
MONTH	USInt		
DAY	USInt		
WEEKDAY	USInt		
HOUR	USInt		
MINUTE	USInt		
SECOND	USInt		
NANOSECOND	UDInt		
SyncVal	Byte		
forVal	Int		
forVal_2	Int		
Value	Byte		
▼ Constant			
CompVal	Byte	16#34	
Value_01	Byte	16#11	
Value_01_DW	DWord	16#A165_D992	
Value_02_DW	DWord	16#58BE_4401	
▼ Return			
MHJ-PLC-Lab-Function-S71200	Void		

```
0001
0002 #Value:=PEEK(area := 16#82,
0003      dbNumber := 0,
0004      byteOffset := 511);
0005 #Value := #Value + 1;
0006
0007 POKE(area := 16#82,
0008      dbNumber := 0,
0009      byteOffset := 511,
0010      value := #Value);
0011
0012 POKE(area:=16#81,
0013      dbNumber:=0,
0014      byteOffset:=1016,
0015      value:=#Value_01_DW);
0016 POKE(area := 16#81,
0017      dbNumber := 0,
0018      byteOffset := 1020,
0019      value := #Value_02_DW);
0020
0021 POKE(area := 16#81,
0022      dbNumber := 0,
0023      byteOffset := 511,
0024      value := B#16#00);
0025
0026 FOR #forVal := 0 TO 120 DO
0027     FOR #forVal_2:=0 TO 10 DO
0028         #rdTimeReturn:=RD_SYS_T(#outputTime);
0029         #rdTimeReturn := WR_SYS_T(#outputTime);
0030         #rdTimeReturn := RD_SYS_T(#outputTime);
0031         #rdTimeReturn := WR_SYS_T(#outputTime);
0032     END_FOR;
0033     #SyncVal:= PEEK(area := 16#81,
0034                     dbNumber := 0,
0035                     byteOffset := 511);
0036     IF #SyncVal = #CompVal THEN
0037        GOTO M_1;
0038     END_IF;
0039 END_FOR;
0040 RETURN;
0041
0042 M_1:
0043 POKE(area := 16#81,
0044      dbNumber := 0,
0045      byteOffset := 511,
```

```
0046      value := B#16#0);
0047
0048
0049
```

Symbol	Address	Type	Comment
#CompVal	16#34	Byte	
#forVal		Int	
#forVal_2		Int	
#outputTime		DTL	
#rdTimeReturn		Int	
#SyncVal		Byte	
#Value		Byte	
#Value_01_DW	16#A165_D992	DWord	
#Value_02_DW	16#58BE_4401	DWord	

Program blocks

Scaling [FC1]

Scaling Properties							
General							
Name	Scaling	**Number**	1	**Type**	FC	**Language**	SCL
Numbering	automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
▼ Input			
SENSOR	Real		
▼ Output			
Scaled_in cm	DInt		
Scaled_Sensor	Real		
InOut			
▼ Temp			
Scaled_in_cm_real	Real		
Constant			
▼ Return			
Scaling	Void		

```
0001 //Convert the range 0.0- 10.0 to Percentage 0-100%
0002 #Scaled_Sensor := 10.0 * #SENSOR;
0003
0004 //Convert the range 0.0- 10.0 to Height range 0-300 cm
0005 #Scaled_in_cm_real := #SENSOR * 30.0;
0006
0007 #"Scaled_in cm" := REAL_TO_INT(#Scaled_in_cm_real);
```

Symbol	Address	Type	Comment
#"Scaled_in cm"		DInt	
#Scaled_in_cm_real		Real	
#Scaled_Sensor		Real	
#SENSOR		Real	

Program blocks

Automation Control (ON/OFF) [FC2]

Automation Control (ON/OFF) Properties							
General							
Name	Automation Control (ON/OFF)	**Number**	2	**Type**	FC	**Language**	SCL
Numbering	automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
▼ Input			
Level in Percentage	Real		
Drain Button	Bool		
▼ Output			
Fill Valve	Real		
Drain Valve	Real		
InOut			
Temp			
Constant			
▼ Return			
Automation Control (ON/OFF)	Void		

```
0001 //IF tank level is < 40% = Fill Valve Opening 100%
0002 //IF tank level is > 40% AND < 80% = Fill Valve Opening 50%
0003 //IF tank level is > 80% = Fill Valve Opening 0%
0004
0005 IF #"Level in Percentage" < 40.0 THEN
0006     #"Fill Valve" := 10.0;  // Statement section IF
0007
0008 ELSIF #"Level in Percentage" > 40.0 AND #"Level in Percentage" < 80.0   THEN
0009     #"Fill Valve" := 5.0; // Statement section ELSIF
0010
0011 ELSIF  #"Level in Percentage" > 80.0 THEN
0012     #"Fill Valve" := 0.0;// Statement section ELSE
0013
0014 END_IF;
0015
0016 //Use Toggle button TO drain the level (at 50% flow rate)
0017 IF #"Drain Button" = 1 THEN
0018     #"Drain Valve" := 5.0;
0019 END_IF;
```

Symbol	Address	Type	Comment
#"Drain Button"		Bool	
#"Drain Valve"		Real	
#"Fill Valve"		Real	
#"Level in Percentage"		Real	

Program blocks

Proportional Control Logic [FC3]

Proportional Control Logic Properties							
General							
Name	Proportional Control Logic	**Number**	3	**Type**	FC	**Language**	SCL
Numbering	automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
▼ Input			
Set Point	Real		
SENSOR	Real		
Kp	Real		Proportional Gain
▼ Output			
Drain Valve	Real		
Filling Valve	Real		
InOut			
▼ Temp			
error	Real		
kpp	Real		proportional gain
u	Real		
Constant			
▼ Return			
Proportional Control Logic	Void		

```
0001 // Input ID38 0-10.0 == 1 to 10
0002 // Scaling Formulae
0003 // Scaled Output = ((Output Max - Output Min)*(Input to be scaled - Input min.))/(Input max - Input min) +
     Output Min.
0004 // In our case:
0005 // Input Min = 0, Input Max = 10, Output min = 1,
0006 // Output Max = 10 and Input to be scaled is Kp (Proportional gain from Pot)
0007 #kpp := (((10.0 - 1.0)/(10.0 - 0.0))*(#Kp - 0.0))+ 1.0;
0008
0009 //Error
0010 #error := #"Set Point" - #SENSOR;
0011
0012 //Proportional Control
0013 #u := #error * #kpp;
0014
0015 //Output
0016 #"Filling Valve" := #u;
0017 #"Drain Valve" := - #u; // SP = 50% and Level is 100%
0018
0019
```

Symbol	Address	Type	Comment
#"Drain Valve"		Real	
#"Filling Valve"		Real	
#"Set Point"		Real	
#error		Real	
#Kp		Real	Proportional Gain
#kpp		Real	proportional gain
#SENSOR		Real	
#u		Real	

Program blocks / System blocks

Program resources

This folder is empty.